

Build the AdHush TV Listener

A Raspberry Pi 4 in a **NESPi 4** case, with a webcam and a microphone, that sits on a shelf, **watches and listens** to your TV, and turns the sound down to a whisper when a commercial starts — then back up when your show comes back. It gives the TV its orders through **one cable** into the TV's serial socket, so it works even when the Wi-Fi is having a bad day. **No HDMI boxes, no relay, no scissors, no soldering.**

NESPi 4 edition · Sharp TV by serial cable · AdHush 0.15

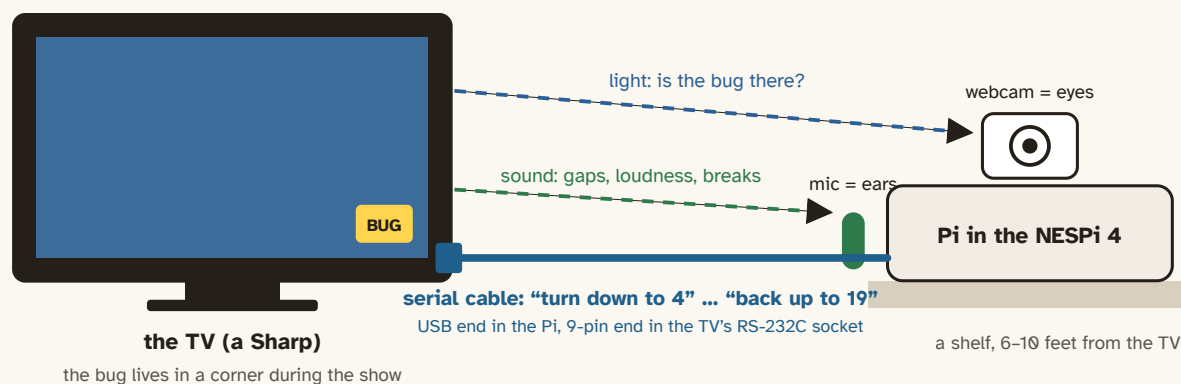
Read this first

Everything here runs on low-voltage USB power. The only thing that plugs into a wall outlet is the Pi's own power brick, and you never open that. Ask an adult for the two steps marked **ADULT CHECK**. Typing must be exact: capital letters and spaces matter.

Time: an afternoon. **Difficulty:** if you can follow a recipe and type carefully, you can do this.

The big picture

The webcam looks at the TV and watches the little channel logo in the corner. TV people call that logo the **bug**, because it sits there all through the show like a bug on the glass. During a commercial the bug is taken off the screen. The microphone listens for the quiet gaps and the loudness jumps around commercials, and remembers the sound of every break you teach it. When the Pi is sure, it tells the TV down the cable: *turn down to 4*. When the show is back: *turn back up*.

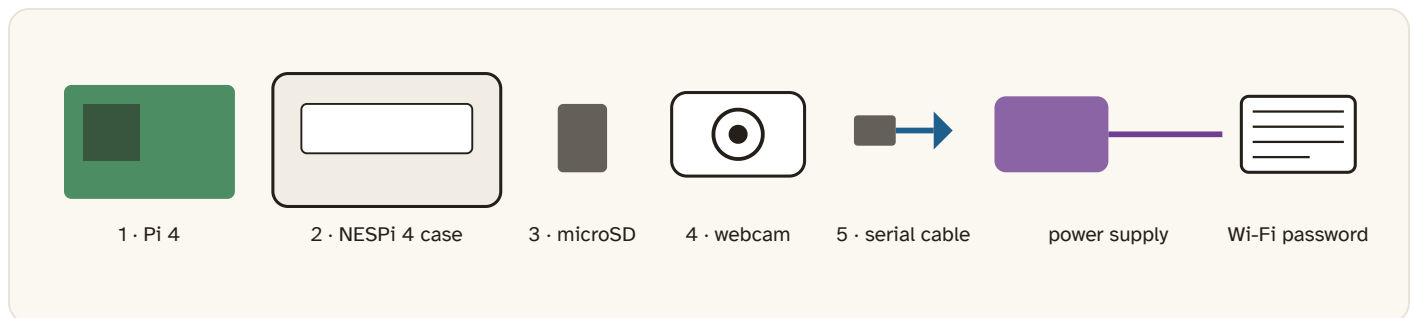


One cable carries the orders. Wi-Fi is still used for the phone page, but the TV never depends on it.

• The parts (about \$125–160, less if you already own a Pi)

#	Part	What it does	Rough price
1	Raspberry Pi 4 (2 GB or more) and the official USB-C power supply	The brain	\$45–60
2	Retroflag NESPi 4 case (comes with a fan, heatsinks and screws)	The body — and it looks great on a shelf	\$30–40
3	microSD card, 32 GB	The brain's memory	\$8
4	USB webcam, 1080p, with a built-in microphone (most have one)	The eyes and the ears	\$20–30
5	USB-to-serial null-modem cable with an FTDI chip: a USB plug on one end, a 9-pin female plug on the other, marked “null modem” or “crossover”; long enough to reach from the shelf to the back of the TV	The voice: how the Pi tells the TV to turn down	\$15–20
6	<i>Optional:</i> a small USB microphone (and a small USB hub, since the case has two ports), if the webcam has none or hears badly	Better ears	\$10–15
7	A shelf, table or small tripod 6–10 feet in front of the TV	Somewhere to stand	—

Just for setup, borrow: a keyboard, a mouse, a monitor or TV with HDMI (and a micro-HDMI cable for the Pi), and your home Wi-Fi password (the Pi still uses Wi-Fi for the phone page, not for the TV). The TV must be a Sharp AQUOS with an **RS-232C** socket on the back: 9 pins, D-shaped, **male** (pins, not holes). The cable end that goes there must be **female** and **cross-wired** (“null modem”). A plain straight serial cable looks the same and does nothing — the one mistake this build can make.



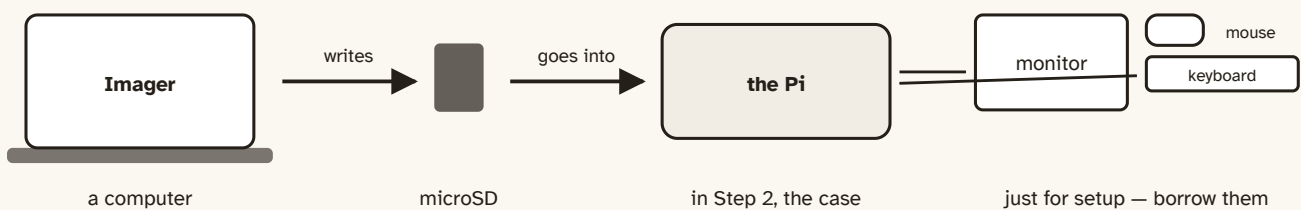
1 Wake up the Pi ADULT CHECK

1. On a computer, install **Raspberry Pi Imager** from raspberrypi.com. Put in the microSD card.
2. Choose **Raspberry Pi OS (64-bit)** — the plain, recommended one at the top of the list. **Not** the one called *Legacy* (its Python is too old) and **not** *Lite* (no desktop, so no Terminal window to type in).
3. In the imager's settings gear, set a username and password and type in your Wi-Fi name and password. Write the card.
4. Put the card in the Pi. Plug in the keyboard, mouse, monitor and the webcam, then the power. Wait for the desktop.
5. Open the black **Terminal** window and type these four lines, pressing Enter after each:

```
git clone https://github.com/socrtwo/adhush
cd adhush
sudo scripts/install-pi.sh
cp config/adhush-listener.example.toml config/adhush.toml
```

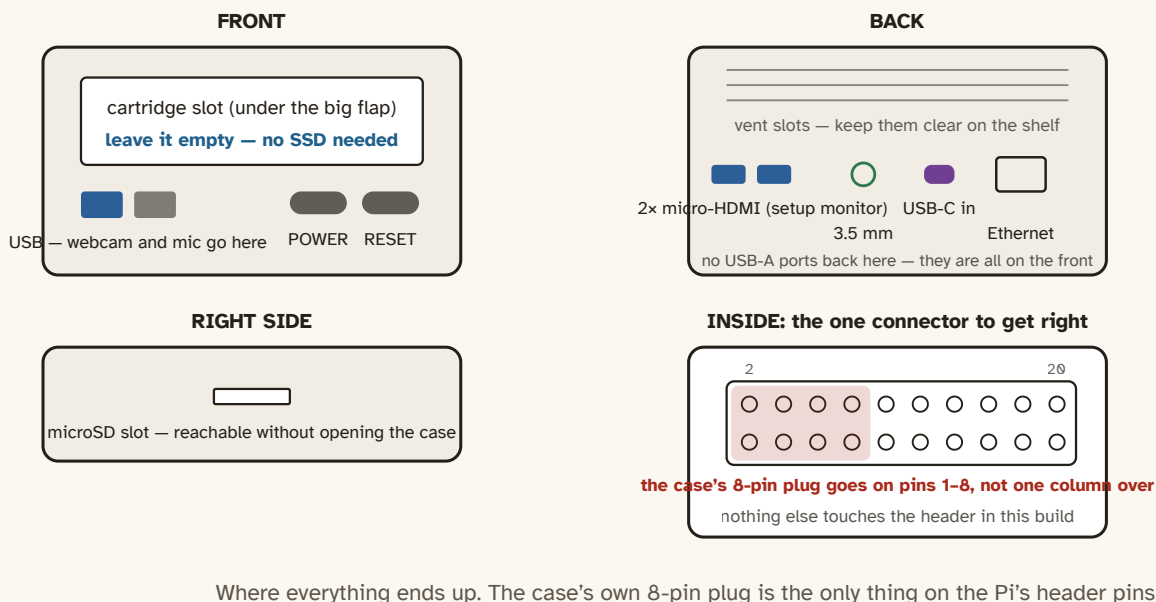
What that last line did

It made **your settings file**, `config/adhush.toml`. It stays on the Pi and is never sent anywhere, which matters because the TV's password will go in it later. You will open it twice, in Steps 3 and 4, with the little text editor `nano`. In nano, **Ctrl+O** then Enter saves, and **Ctrl+X** leaves.



2 Build the Pi into the NESPi 4 ADULT CHECK

The case comes with its own little manual and screws; follow that for where each screw goes. Here is what matters for the listener. Good news: unlike the older AdHush box, **nothing leaves the case** — no wires, no relay — so the lid can be screwed down as soon as it is built.



1. Stick the **heatsinks** from the case box onto the Pi's two biggest chips. The listener watches and listens all evening, so they matter.

- Follow the case manual to screw the Pi in and plug its USB and Ethernet ports into the case's **adapter board**. Plug the **fan** into the fan header on the case board.
- Push the case's **8-pin plug** onto header pins **1–8**: the two rows of four at the corner nearest the microSD slot. The manual shows which way round. **Count the pins**: it must not sit one column over.
- Set the case board's **SAFE SHUTDOWN** switch to **OFF**. The ON position needs a helper script that no longer works on current Raspberry Pi OS. OFF turns the POWER button into a plain power switch — fine for a box that stays on.
- Leave the **cartridge empty** for now. AdHush needs very little storage; if you want to use the slot for an SSD anyway, the next page shows how — after the box works from the card.
- Close the case and screw the lid down. Put the microSD card in the slot on the right side.

2½

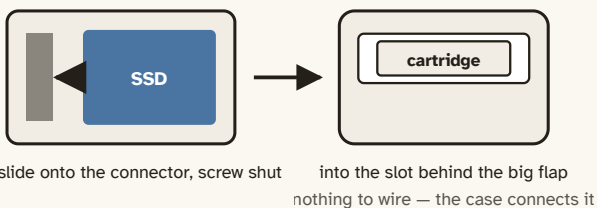
Optional: an SSD in the cartridge ADULT CHECK

The NESPi 4's front slot takes a **cartridge** that holds a 2.5-inch SATA SSD. AdHush does not need one — it stores only numbers, and a microSD card has room for years of them. But an SSD boots faster, shrugs off the thousands of tiny writes a box makes over a year, and is the part most likely to survive the POWER button cutting the power mid-write. If you have one, this is how to use it. **Do it after Step 1 works from the card**, so you always have a card that boots.

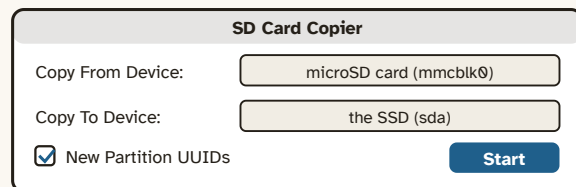
You need

A 2.5-inch **SATA** SSD, 7 mm thick — the normal laptop kind; 120 GB is plenty. **Not** an M.2 or NVMe stick: those do not fit the cartridge. The cartridge shell comes in the case box.

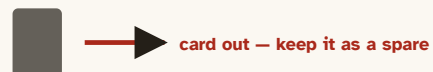
1 • SSD into the cartridge, cartridge into the slot



2 • copy the card onto the SSD, then take the card out



then `sudo raspi-config` → Advanced Options → Boot Order → NVMe/USB Boot
`sudo poweroff`, POWER off, card out, POWER on



- Put the SSD in the cartridge.** Open the cartridge shell (the small screws), slide the SSD onto the connector inside until it clicks, and screw the shell shut. Push the cartridge into the slot behind the big front flap.
- Copy the card onto the SSD.** With the Pi running from the microSD card, open the menu → **Accessories** → **SD Card Copier**. *Copy From*: the microSD card. *Copy To*: the SSD (it shows up as a USB drive with the SSD's name). Tick **New Partition UUIDs**, press **Start**, wait a few minutes. Everything comes along — AdHush, your settings file, and any breaks it has learned.

3. **Tell the Pi to boot from USB.** In the Terminal, `sudo raspi-config` → **Advanced Options** → **Boot Order** → **NVMe/USB Boot**, then Finish. If it offers to update the bootloader, say yes (Pi 4 boards made before late 2020 need that once).
4. **Switch over.** Type `sudo poweroff`, wait for the green light to stop blinking, press POWER, take the microSD card out of the slot on the right side, and press POWER again. The Pi now starts from the SSD. Keep the card: it is your spare.
5. **Check.** Type `findmnt /` — it should say `/dev/sda2`, not `/dev/mmcblk0p2`. Then carry on with Step 3.

Two things to know about the SSD

Wi-Fi. The cartridge uses a USB 3.0 link, and USB 3.0 can put noise on 2.4 GHz Wi-Fi. The TV is on a cable, so it does not care; if the *phone page* becomes hard to reach after adding the SSD, join your router's 5 GHz network, or plug an Ethernet cable into the back of the case.

Switching off. With SAFE SHUTDOWN off, the POWER button cuts the power like pulling the plug. The listener is meant to stay on, but when you do want it off, type `sudo poweroff` first and press POWER once the green light has stopped blinking.

If it still boots from the card: take the card out and re-check Boot Order. **Rainbow screen or “no boot device”:** reseal the cartridge, then the SSD inside it; put the card back to prove the Pi is fine. **Very slow or freezes:** a few USB-to-SATA bridges misbehave with the Pi 4 — an adult can search “Raspberry Pi 4 usb-storage quirks” for the one-line fix in `/boot/firmware/cmdline.txt`.

3 Plug in the eyes and ears

Plug the **webcam** into one of the two USB ports under the case's small front flap. The other port is for the serial cable (Step 4). If you also want a separate USB microphone, put a small USB hub in one of the ports. Then ask the Pi what it found:

```
adhush doctor
```

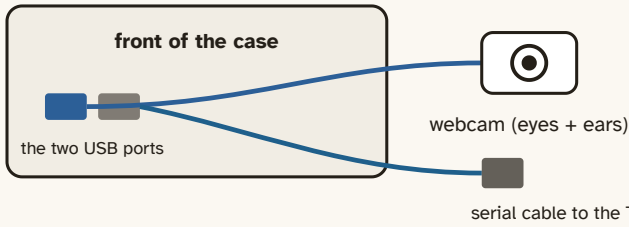
It prints a list of **video devices** (the webcam is usually `/dev/video0`) and a list of **sound cards** with numbers, something like `card 1: Webcam`. Find the line for your microphone and remember its number. Now open your settings file:

```
nano config/adhush.toml
```

Near the top, check these two lines. Change the **1** to your microphone's card number if it is different:

```
device = "/dev/video0"           # the webcam
audio_device = "alsa:plughw:1,0"  # the microphone: card 1, device 0
```

Save with **Ctrl+O**, Enter, then leave with **Ctrl+X**.



```
$ adhush doctor
video devices: /dev/video0
sound cards:
card 0: Headphones
card 1: Webcam <- this one
```

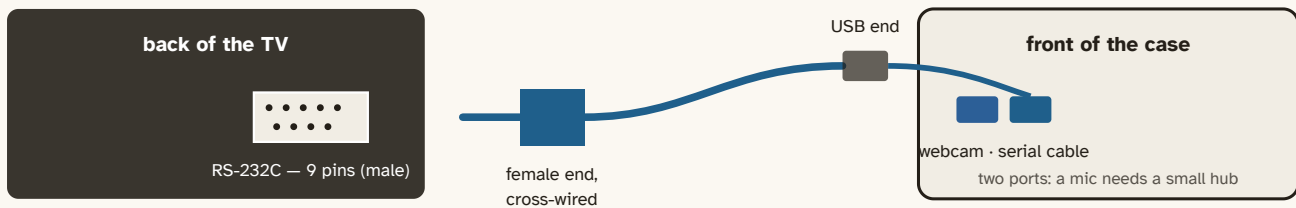
The number after “card” is what goes in the settings file.

4

Let the TV take orders

ADULT CHECK

Plug the **9-pin end** of the serial cable into the **RS-232C** socket on the back of the TV (it only fits one way; tighten the two thumbscrews if it has them). Plug the **USB end** into the free front USB port on the case. That is the whole wiring.



a plain straight serial cable looks identical and does nothing — it must say null modem / crossover

Now ask the Pi where the cable landed, and put yourself on the list of people allowed to use serial ports (Linux keeps one); the reboot makes it take:

```
ls /dev/ttyUSB*          # should answer /dev/ttyUSB0
sudo usermod -aG dialout $USER
sudo reboot
```

After the reboot, open the settings file (`nano config/adhush.toml`) and check the `[control]` part. It already says the right things; make sure the port matches what `ls` showed:

```
[control]
backend = "rs232_sharp"

[control.rs232_sharp]
port = "/dev/ttyUSB0"    # what ls /dev/ttyUSB* showed
baud = 9600
duck_level = 4           # how quiet a commercial gets (4 is a whisper)
```

Save (**Ctrl+O**, Enter, **Ctrl+X**). Now test it — this is the exact thing the listener does when a commercial comes on:

```
adhush probe --active # asks the volume, turns the TV down to 4, waits, turns it back up
```

What you should hear – and what the messages mean

The sound dips for two seconds and comes back: the TV path works. That is the whole point of this step.

“cannot open serial port” / “Permission denied”: the `dialout` line has not taken yet — reboot.

“serial port /dev/ttyUSB0 not present”: the USB end is not in the Pi, or it landed on `ttyUSB1` — fix the port line.

“timeout” (the TV says nothing): a straight cable instead of null-modem, or the plug is not in the RS-232C socket. The TV must be on; standby is fine.

“rejected VOLM” / ERR: wrong socket, or a second control cable is fighting it.

Rather use Wi-Fi?

The TV takes the same orders over the network: MENU → Initial Setup → Internet Setup → Network Setup → IP Control Setup → Enable. Set `backend = "network_ip"` and fill in the `[control.network_ip]` part of the settings file with the TV's address and login instead. The phone app uses that path; the Sharp allows only one such connection at a time, so the app must be stopped while the Pi is running. The cable has no such limit and no password.

5 Point the eyes

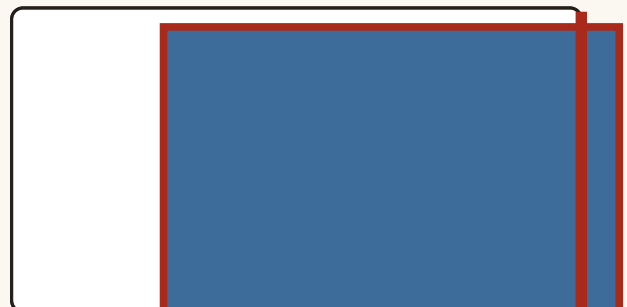
Put the case and webcam on the shelf, six to ten feet from the TV, roughly straight in front of it. The **whole TV must be in the webcam's picture** with a little room around it. This is a rule the listener never breaks: if the TV runs off the edge of the picture, it will not use the camera at all. It never guesses from half a TV.

GOOD — whole TV, room around it



the camera sees the bug's corner → it can help

NOT GOOD — the TV runs off the edge



“whole TV not in view” → the camera stays out of it

The phone page (Step 6) tells you which of these the camera sees: “bug seen”, “looking for the bug”, or “whole TV not in view”.

Now, with a **show** on — not a commercial — teach it what the channel's bug looks like:

```
adhush calibrate          # run this while a SHOW is on, not an ad
```

It watches for twenty seconds and saves the bug. If a commercial came on in the middle, just run it again. Set-up is once per channel.

If the bug is not in the bottom-right corner

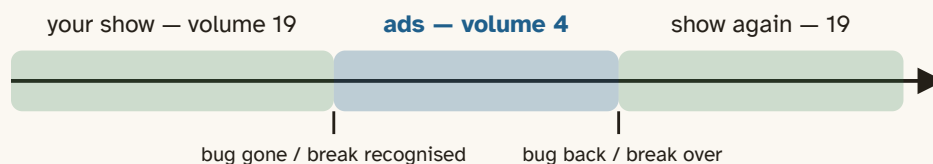
The listener expects the bug in the **bottom-right**. If your channel keeps it somewhere else, open the settings file and find the `roi` line under `[detect.logo_absence]`. `x` and `y` say how far across and down the corner is, from 0 to 1; `w` and `h` are its size. Bottom-left would be `x = 0.02, y = 0.80`; top-right would be `x = 0.84, y = 0.06`. Then run `adhush calibrate` again.

6 Wake it up

```
adhush run                # start watching and listening!
```

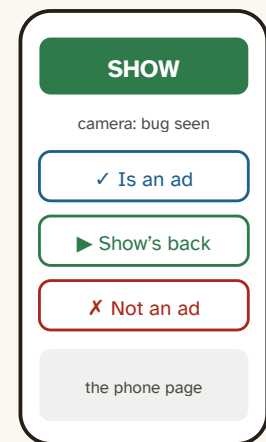
Watch TV. When a commercial break starts, the sound drops to a whisper; when the show returns, it comes back. The first day it mostly listens and learns. It gets noticeably smarter about breaks it has heard before.

On your phone: open a browser and go to `http://` + the Pi's address + `:8675` (the Pi prints its address when it starts; `hostname -I` in the Terminal shows it too). The page shows SHOW, MUTED or TEACHING live, what the camera can see, and three buttons that make the listener smart:



slow to turn down, quick to turn back up — on purpose.
Missing the first second of an ad is fine.
Missing the first second of your show is not.

It never mutes: at volume 4 the microphone still hears the TV,
which is how it knows the show is back.



- ✓ **Is an ad** — press the moment a commercial break starts. The TV turns down and stays down.
- ► **Show's back** — press the second your show returns. The sound comes back and everything in between is remembered as a break (as numbers, never a recording).
- ✗ **Not an ad** — press if it turned your show down by mistake. The sound comes back at once, whatever caused it is dropped, the camera has to see the bug again before it may say the bug is gone, and for one minute nothing may turn the TV down.

Do the ✓ / ► pair for five or six breaks. After that it recognises those commercials by itself within a few seconds.

Make it start by itself

Once it works, type `sudo systemctl enable --now adhush`. From then on the listener starts whenever the case is switched on, and if it ever loses power while the TV is turned down, it puts the volume back the next time it starts.

How it decides

The listener uses four ways of spotting a commercial. Two are hints that need a second opinion; two are strong enough to turn the TV down on their own.

Method	What it uses	Can turn the TV down alone?
Quiet gaps	microphone: the short silence broadcasters leave between the show and each ad	no — needs a second opinion
Loudness jumps	microphone: the sound jumping up and staying up (ads are mixed louder)	no — needs a second opinion
Remembered breaks	microphone: fingerprints of the breaks you taught it with ✓ and ►	yes
Channel bug	webcam: the logo corner — only while the whole TV is in view and the bug has been seen at least once	yes

The phone app has two more (spoken words and on-screen captions); they are not on the Pi yet. Because the Pi stands still on a shelf, the camera method is steadier here than in a hand.

Words used in this guide

- **Bug** — the small channel logo in a corner of the picture during a show. It is taken off the screen during commercials.
- **Duck / turn down** — the TV goes to volume 4 instead of muting, so the microphone can still hear when the show is back.
- **Remembered break** — the sound of a break you bracketed with ✓ and ►, stored as numbers (never a recording).
- **Settings file** — `config/adhush.toml`, the one file you edit. It stays on the Pi.

? If something's wrong

Problem	Try
<code>adhush probe --active</code> says <i>cannot open serial port</i> or <i>Permission denied</i>	Run the <code>dialogout</code> line from Step 4 and reboot

Problem	Try
It says <i>serial port ... not present</i>	The USB end is not in the Pi, or it is <code>ttyUSB1</code> : check <code>ls /dev/ttyUSB*</code> and the port line
It says <i>timeout</i> — the TV says nothing	Straight cable instead of null-modem, or not in the RS-232C socket. The TV must be on
Using Wi-Fi instead and it says <i>hung up before asking for a login</i>	Something else is connected — press Stop in the phone app and try again
The phone page says <i>camera: whole TV not in view</i>	Move the shelf back or turn the webcam until the whole TV, with a border, is in the picture (Step 5)
The phone page says <i>camera: looking for the bug for minutes</i>	Run <code>adhush calibrate</code> again during a show, or fix the corner in the settings file (Step 5)
It turns the show down by mistake	Press X Not an ad . If it keeps happening on one channel, calibrate again
It never turns anything down	Teach it five breaks with ✓ / ►. Check the microphone card number (Step 3)
Sound stuck quiet after the Pi lost power	Switch the case back on: the listener puts the volume back by itself. Or use the remote
The case gets hot	Make sure the fan is plugged in (Step 2) and the vent slots on the back are not against a wall
<code>install-pi.sh</code> says the Python version is too old	The <i>Legacy</i> image got written. Re-flash with Raspberry Pi OS (64-bit), Step 1
<code>adhush run</code> errors	Read the message; <code>adhush doctor</code> lists what is missing

AdHush is free software (MIT). This guide matches AdHush 0.15 and `config/adhush-listener.example.toml`; the same steps in plain text are in `docs/build-guide-tv-listener.md`. Nothing you build here records sound or pictures — the listener keeps only numbers about them, and nothing leaves the Pi except the “turn down” and “turn up” orders down the cable to the TV.